

Failover cluster env with WS 2019 with SQL 2019 and in place upgrade to WS 2025 (Datacenter edition) and SQL 2022.**Query:**

We have a failover cluster env with WS 2019 with SQL 2019 and planning to place upgrade to WS 2025 (Datacenter edition) and SQL 2022. This should be rolling upgrade OS first then SQL in-place upgrade on each node. Any issues with this approach or anyone guide to find a step-by-step document to accomplish this.

Short answer: Your plan (OS first, then SQL) is solid **but you can't roll a WSFC directly from Windows Server 2019 → 2025**. Cluster OS Rolling Upgrade only supports **adjacent** versions. So you must either (A) go 2019→2022→2025 in two passes, or (B) build a new 2025 cluster and migrate to it. SQL Server 2019 is supported on Windows Server 2025, so OS-first is fine from a supportability standpoint. ([Microsoft Learn](#))

Two supported paths**A) Sequential rolling OS upgrade (stay on same cluster)**

1. Roll 2019 → 2022 across nodes (mixed-OS cluster during the window).
2. Raise the cluster functional level.
3. Roll 2022 → 2025 across nodes.
4. Raise the cluster functional level again.

Then in-place upgrade SQL 2019 → 2022. ([Microsoft Learn](#))

B) Side-by-side migration (often less risk)

Stand up a new WS 2025 cluster, then:

- If you use **AGs**: add 2025/SQL 2022 replicas, seed, sync, fail over, then decommission old replicas. Minimal downtime. ([Microsoft Learn](#))
- If you use **FCI**: deploy a new FCI on 2025 (new disks or planned storage move), restore databases/logs, cut over the listener/VNN at a maintenance window. (Microsoft's FCI upgrade guide covers the moving parts.) ([Microsoft Learn](#))

High-level runbook (condensed)**Pre-flight (before any OS upgrade)**

- Patch Windows & SQL to current CUs; verify backups, scripts, Agent jobs, SSIS packages, DTC usage, antivirus exclusions.
- Validate the cluster:
- Test-Cluster -Node NODE1,NODE2 | Out-File C:\Temp\ClusterValidation.html

Fix any red flags before proceeding. ([Microsoft Learn](#))

OS rolling upgrade (do this twice: 2019→2022, then 2022→2025)**For each node in turn:**

On the node you're about to upgrade

Suspend-ClusterNode -Drain

Perform the in-place OS upgrade to the next version

(2019→2022 first; later 2022→2025)

Reboot / complete setup, confirm node rejoins cluster

Resume-ClusterNode

After **all** nodes are on the new OS version:

Update-ClusterFunctionalLevel # one-way change; do this only when stable

Repeat the cycle for the next hop. (You cannot skip directly to 2025.) ([Microsoft Learn](#))

Notes: Mixed-version clusters are meant to be temporary—finish each hop and raise the functional level promptly. ([Microsoft Learn](#))

SQL Server upgrade to 2022

If you run an FCI:

- Follow Microsoft's "Upgrade a SQL Server failover cluster instance" guide. In brief: run Setup on the active node to upgrade the instance; expect a brief outage when the engine is upgraded; then run Setup on the remaining nodes to add them back at the new version; finally apply the latest SQL 2022 CU across all nodes. ([Microsoft Learn](#))

If you run AGs:

- Do a **rolling SQL upgrade**: upgrade secondaries to SQL 2022 → fail over to a 2022 secondary → upgrade the former primary last. Keep database compatibility at 150 initially; raise to 160 after testing. ([Microsoft Learn](#))

Gotchas to watch

- **Support matrix:** Windows Server 2025 supports SQL 2019 and SQL 2022, so OS-first is supported. ([Microsoft Learn](#))
- **.NET:** SQL 2022 needs .NET Framework **4.7.2** on each node. ([Microsoft Learn](#))
- **Storage/HBAs/MPIO:** ensure 2025-compatible drivers/firmware for your shared storage before upgrading. (Run Test-Cluster storage tests offline.) ([Microsoft Learn](#))
- **Quorum:** double-check witness and node votes before draining nodes so you maintain quorum during each hop. (Cluster doc covers mixed-mode behavior and functional level.) ([Microsoft Learn](#))

Microsoft docs you can follow step-by-step

- **Cluster OS Rolling Upgrade (limitations & process).** You can only upgrade to the **next** OS version per hop. ([Microsoft Learn](#))
- **Update-ClusterFunctionalLevel** (when to raise it). ([Microsoft Learn](#))
- **SQL Server FCI in-place upgrade guide (to 2022).** ([Microsoft Learn](#))
- **Always On AG rolling upgrade guide (to 2022).** ([Microsoft Learn](#))
- **SQL on Windows version support table** (confirms SQL 2019/2022 on WS 2025). ([Microsoft Learn](#))
- **SQL 2022 requirements (.NET 4.7.2, OS).** ([Microsoft Learn](#))

If you are running **both FCI + AG** in the same WSFC, the upgrade plan has to respect both technologies.

Here's how you can structure it safely:

◆ Key Principles

- **OS upgrade comes before SQL upgrade** (2019 → 2022 → 2025).
- **FCI upgrade = in-place, instance-level upgrade** (requires downtime).
- **AG upgrade = rolling, replica-level upgrade** (can be near-zero downtime if done carefully).
- **Mixed OS versions are temporary** — you must finish each hop (2019→2022, 2022→2025) and raise the cluster functional level before the next hop.

◆ High-Level Step-by-Step Runbook

Phase 1 – Preparation

1. Backup everything (system DBs, user DBs, Agent jobs, SSIS, Linked Servers, credentials, keys, certs, etc.).
2. Patch SQL 2019 to the latest CU (Microsoft requires latest patch before upgrade).
3. Validate cluster health:
4. Test-Cluster -Node NODE1, NODE2, NODE3 | Out-File C:\Temp\ClusterValidation.html
5. Check quorum witness & voting — ensure you won't lose quorum during node drains.
6. Confirm storage drivers/HBA firmware are certified for Windows Server 2022/2025.
7. Make sure .NET Framework **4.7.2 or later** is installed.

Phase 2 – Cluster OS Upgrade (Rolling, 2 hops)

Repeat the following cycle **twice** (2019→2022, then 2022→2025):

For each node:

Suspend-ClusterNode -Drain

Perform in-place OS upgrade (2019→2022 first)

Reboot, install patches, validate

Resume-ClusterNode

After all nodes are upgraded in a hop:

Update-ClusterFunctionalLevel

➡ Now the entire WSFC is on Windows Server 2025, functional level raised.
SQL is still 2019.

Phase 3 – SQL Server Upgrade

A) FCI Instances

1. Pick a maintenance window (downtime is unavoidable).
2. On the active node of the FCI, run SQL 2022 Setup → **Upgrade SQL Server Failover Cluster Instance**.
3. This upgrades the binaries + system DBs on that node.
4. Fail over to a passive node → run Setup there to "Add Node" with SQL 2022 binaries.
5. Repeat for remaining nodes in the FCI.
6. Apply the latest SQL 2022 CU across all nodes.

B) Availability Groups

(Do this after all FCIs hosting AG replicas are upgraded.)

1. Upgrade **secondary replicas first**:
 - On each replica node, run in-place upgrade SQL 2019 → SQL 2022.
 - Restart SQL service, rejoin replica to AG.

2. Fail over AG to a SQL 2022 secondary.
3. Upgrade the **old primary** last.
4. Keep DB compatibility level at 150 (SQL 2019) initially; raise to 160 later after regression testing.

Phase 4 – Post-Upgrade Tasks

- Validate failovers for both **FCI** and **AG**.
- Run DBCC CHECKDB on all user DBs.
- Validate jobs, linked servers, replication, SSRS/SSIS (if applicable).
- Monitor error logs for upgrade warnings.
- Document new baseline performance.

◆ Downtime vs No-Downtime

- **FCI**: requires downtime during the instance upgrade step.
- **AG**: can be done with rolling upgrade, downtime limited to failover time.

✓ Bottom line:

- Do OS rolling upgrade (2019→2022→2025).
- Upgrade FCIs in-place (downtime required).
- Then perform AG rolling upgrade to SQL 2022.
- Validate, patch, test.

Runbook for upgrading a Windows Server Failover Cluster (WSFC) running both FCI + AG from WS 2019 + SQL 2019 → WS 2025 + SQL 2022.

SQL Server FCI + AG Cluster Upgrade Runbook From WS 2019 + SQL 2019 → WS 2025 + SQL 2022

1 Pre-Upgrade Preparation

- ✓ Take **full backups**: system DBs, user DBs, Agent jobs, SSIS, certificates, keys, linked servers.
- ✓ Apply **latest CU** to SQL 2019.
- ✓ Confirm cluster health:
- Test-Cluster -Node NODE1, NODE2, NODE3 | Out-File C:\Temp\ClusterValidation.html
- ✓ Verify quorum witness & votes (avoid losing quorum when draining nodes).
- ✓ Ensure .NET Framework **4.7.2+** installed.
- ✓ Validate storage drivers/firmware are supported on WS 2022/2025.

2 OS Rolling Upgrade (Two Hops: 2019→2022 → 2025)

Repeat per node (for both hops):

Suspend-ClusterNode -Drain

Perform in-place OS upgrade (2019→2022 first)

Reboot, apply patches

Resume-ClusterNode

After all nodes upgraded for a hop:

Update-ClusterFunctionalLevel

→ End of Phase 2: All nodes now WS 2025, cluster functional level raised. SQL still at 2019.

3 SQL Server Upgrade

A) Failover Cluster Instance (FCI)

1. Schedule **downtime** (all DBs in FCI offline during upgrade).
2. On active node → Run **SQL 2022 Setup → Upgrade SQL Server Failover Cluster Instance**.
3. Complete upgrade, verify DBs online.
4. Failover FCI → passive node.
5. On passive node → Run Setup → **Add Node** (upgrades binaries).
6. Repeat for remaining nodes.
7. Apply latest SQL 2022 CU on all FCI nodes.

B) Availability Groups (AGs)

1. Upgrade **secondary replicas first** (in-place SQL 2019 → SQL 2022).
2. Restart services, rejoin replicas to AG.
3. Fail over AG to a SQL 2022 secondary.
4. Upgrade the **former primary** last.
5. Keep DB compatibility level at 150 initially. Raise to 160 after testing.

4 Post-Upgrade Validation

- ✓ Validate **FCI failover** across nodes.
- ✓ Validate **AG failover** across replicas.
- ✓ Run DBCC CHECKDB on all user DBs.
- ✓ Test Agent jobs, linked servers, SSIS/replication.

-  Monitor SQL & cluster logs for warnings.
-  Document new baseline (PerfMon, Query Store).

Rollback Plan

- If OS upgrade fails → evict node, reinstall WS 2019, rejoin cluster.
- If SQL upgrade fails → restore full backups to a standby SQL 2019 instance, rejoin to AG, cut back.
- Always keep **one node un-upgraded** until others are validated.

Notes:

- FCI = downtime required.
- AG = rolling upgrade with minimal downtime.
- Each OS hop (2019→2022, 2022→2025) must complete across *all* nodes before raising cluster functional level.